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Archives

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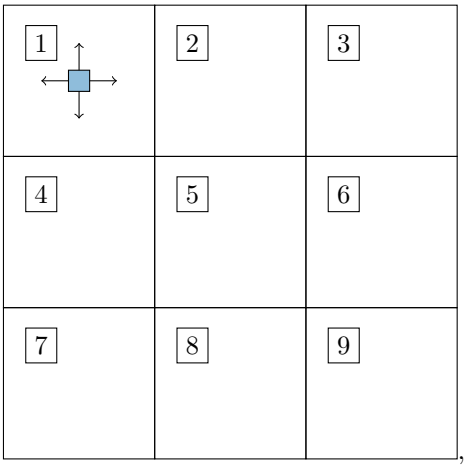
[Reinforcement Learning Theory]$^{2}_{\mathrm{T}}$

Reinforcement Learning Theory

Adapted from *Neuro-Dynamic Programming* by
Dmitri P. Bertsekas and John N. Tsitsiklis.

Motivating example.

Consider the following "environment":



,

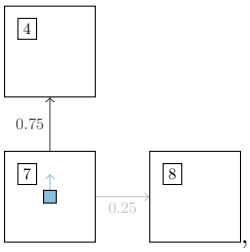
where there are a total of 9 "states" (▫), an "agent" (■), and a set of "actions" (→) that the agent can take at any state.

At every "step" in time, the agent is in one of the states, and must take one of the actions.

At any step, if the agent takes an action in a state, it will be in a state at the next step according to a "transition distribution" (that only depends on the current state and action, not the current step).

For example, the agent could

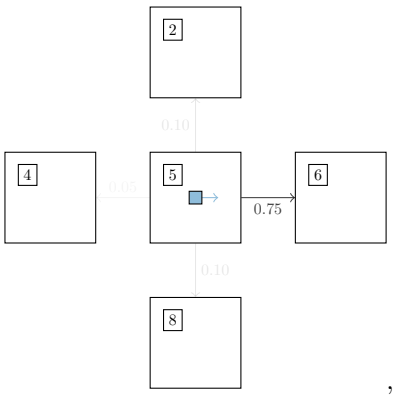
– take action ↑ at state 7:



,

and

– take action → at state 5:



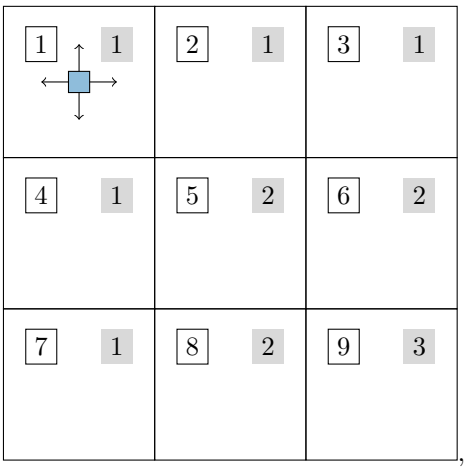
,

showing the probability of being in another state at the next step according to the transition distribution.

However, this begs the question, what state is the agent at initially (step 1)? To decide this, there is also an initial distribution over the states.

Now, let's make things a bit more interesting.

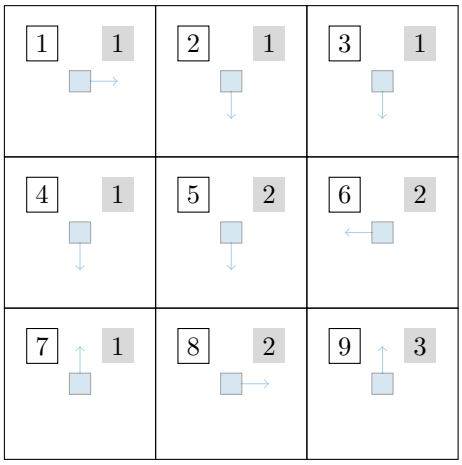
Let every state also have a return that the agent recieves from it, leaving our environment looking something like:

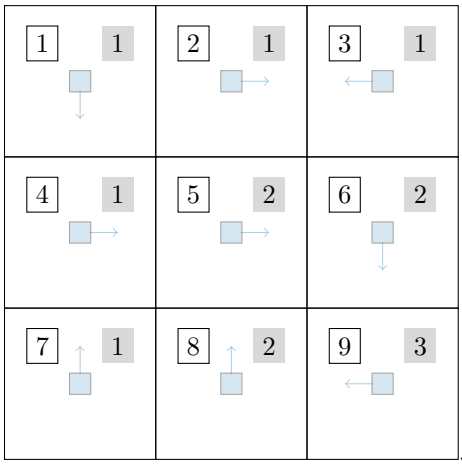


with the returns (·) shown.

Also, let there be some fixed "step limit" such that, when the current step reaches it, everything ends.

Now, suppose you can "program" the agent as follows: for every state and step, you can fix the action that the agent takes. For example, here's what you might program the agent to do at every state for two different steps:





We'll call each such programming a "strategy".

We know that the agent starts in some state according to the initial distribution. After this, we follow the strategy, transitioning states according to the transition distribution, recieving returns at every state until we reach the step limit. Let's refer to this process as a "run", and the sum of the returns over all a run's states as the "total return".

Because there is some initial distribution and transition distribution that are fixed over runs, we know there must be some average total return for every strategy.

Now, consider the following question: what strategy has the greatest average total return?

Notation.

To answer this question, let's establish some notation.

Let's [define]$^{26}_{\mathrm{D}}$

$$\mathbf{S} \coloneqq \text{the set of states},$$

where $\mathbf{S}$ is finite, and [define]$^{30}_{\mathrm{D}}$

$$\mathbf{A} \coloneqq \text{the set of actions},$$

where $\mathbf{A}$ is finite. [Define]$^{19}_{\mathrm{D}}$

$$[R : \mathbf{S} \to \mathbb{R}]^{17}_{\mathrm{N}} \coloneqq \text{mapping from states to returns}.$$

[Define]$^{21}_{\mathrm{D}}$

$$T \coloneqq \text{the step limit, where } T \geq 1,$$

[define]$^{22}_{\mathrm{D}}$

$$\pi = (\pi_1, \ldots, \pi_{T-1}) \coloneqq \text{a strategy, where each "step-strategy" } \pi_t : \mathbf{S} \to \mathbf{A},$$

and [define]$^{34}_{\mathrm{D}}$

$$\mathbf{\Pi} \coloneqq \text{the set of all strategies } \pi.$$

[Note that]$^{4}_{\mathrm{P}}$ the total number of strategies can be written as

$$|\mathbf{\Pi}| = T \times |\mathbf{S}| \times |\mathbf{A}|.$$

[Define]$^{23}_{\mathrm{D}}$ for any $\mathrm{s} \in \mathbf{S}$,

$$[\mathrm{P_I(s)}]^{13}_{\mathrm{N}} \coloneqq \text{the initial distribution probability of starting in state s.}$$

[Define]$^{28}_{\mathrm{D}}$ for any $\mathrm{s} \in \mathbf{S}$, $\mathrm{a} \in \mathbf{A}$, $\pi \in \mathbf{\Pi}$, $1 \leq t \leq T-1$,

$$\mathrm{P_A}\left(\pi, t : \mathrm{a} \mid \mathrm{s}\right) \coloneqq \begin{cases} 1 & \pi_t(\mathrm{s}) = \mathrm{a} \\ 0 & \pi_t(\mathrm{s}) \neq \mathrm{a}. \end{cases}$$

[Define]$^{31}_{\mathrm{D}}$ for any $\mathrm{s}, \mathrm{s}' \in \mathbf{S}$, $\mathrm{a} \in \mathbf{A}$,

$$\mathrm{P_T}\left(\mathrm{s} \mid \mathrm{a}, \mathrm{s}'\right) \coloneqq \text{the transition distribution probability of going to state s after action a in state s}'.$$

[Define]$^{29}_{\mathrm{D}}$ for all $1 \leq t \leq T$,

$$\mathrm{S}_t \coloneqq \text{random variable for the state at step } t,$$

and [define]$^{27}_{\mathrm{D}}$ for all $1 \leq t \leq T-1$,

$$\mathrm{A}_t \coloneqq \text{random variable for the action at step } t,$$

whose distributions are dependent on π.

[Let]$^{20}_{\mathrm{D}}$

$$\mathrm{Pr}_\pi(\cdot) \coloneqq \text{the probability of an event when the agent follows a strategy } \pi \in \mathbf{\Pi},$$

and [let]$^{32}_{\mathrm{D}}$

$$\mathbb{E}_\pi[\cdot] \coloneqq \text{the expectation of a quantity when the agent follows a strategy } \pi \in \mathbf{\Pi}.$$

Because we follow a fixed strategy, [we know that]$^{25}_{\mathrm{D}}$, for all $\pi \in \mathbf{\Pi}$, $\mathrm{s} \in \mathbf{S}$, $\mathrm{a} \in \mathbf{A}$, $1 \leq t \leq T-1$, when defined,

$$\mathrm{Pr}_\pi(\mathrm{A}_t = \mathrm{a} \mid \mathrm{S}_t = \mathrm{s}) = \mathrm{P_A}\left(\pi, t : \mathrm{a} \mid \mathrm{s}\right),$$

[from which it follows]$^{3}_{\mathrm{P}}$ that A_t is independent of all previous states and actions given S_t. Notationally, for all $\pi \in \mathbf{\Pi}$, $1 \leq t \leq T-1$, when defined,

$$[\mathrm{Pr}_\pi\left(\mathrm{A}_t \mid \mathrm{S}_t, \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1\right) = \mathrm{Pr}_\pi\left(\mathrm{A}_t \mid \mathrm{S}_t\right)]^{15}_{\mathrm{N}}.$$

[We know that]$^{18}_{\mathrm{D}}$ for all $\pi \in \mathbf{\Pi}$, $\mathrm{s} \in \mathbf{S}$,

$$\mathrm{Pr}_\pi(\mathrm{S}_1 = \mathrm{s}) = \mathrm{P}_\mathrm{I}(\mathrm{s}),$$

and [we know that]$^{33}_{\mathrm{D}}$ for all $\pi \in \mathbf{\Pi}$, $\mathrm{s}, \mathrm{s}' \in \mathbf{S}$, $\mathrm{a} \in \mathbf{A}$, $2 \leq t \leq T$, when defined,

$$\mathrm{Pr}_\pi(\mathrm{S}_t = \mathrm{s} \mid \mathrm{A}_{t-1} = \mathrm{a}, \mathrm{S}_{t-1} = \mathrm{s}') = \mathrm{P}_\mathrm{T}\left(\mathrm{s} \mid \mathrm{a}, \mathrm{s}'\right),$$

[where]$^{24}_{\mathrm{D}}$ S_t is independent of all previous states and actions given A_{t-1} and S_{t-1}. Notationally, for all $\pi \in \mathbf{\Pi}$, $2 \leq t \leq T$, when defined,

$$\mathrm{Pr}_\pi\left(\mathrm{S}_t \mid \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1\right) = \mathrm{Pr}_\pi\left(\mathrm{S}_t \mid \mathrm{A}_{t-1}, \mathrm{S}_{t-1}\right).$$

We can now write the total return as

$$\sum_{t=1}^{T} R(\mathrm{S}_t)$$

and the average total return as

$$\mathbb{E}_\pi\left[\sum_{t=1}^{T} R(\mathrm{S}_t)\right].$$

Preliminaries.

Let's now describe some preliminary results using this notation.

One of the first questions we may ask is, does our notation completely describe the distribution of runs for a particular strategy? That is,

$$\text{for any particular sequence of states } \mathrm{s}_1, \ldots, \mathrm{s}_T \in \mathbf{S} \text{ and actions } \mathrm{a}_1, \ldots, \mathrm{a}_{T-1} \in \mathbf{A},$$

and

$$\text{for any strategy } \pi \in \mathbf{\Pi},$$

can we determine the value of

$$\mathrm{Pr}_\pi(\mathrm{S}_1 = \mathrm{s}_1, \mathrm{A}_1 = \mathrm{a}_1, \mathrm{S}_2 = \mathrm{s}_2, \ldots, \mathrm{S}_T = \mathrm{s}_T)?$$

The answer to this question is: yes! We have all the information we need to completely describe this joint distribution. [It can be written]$^{5}_{\mathrm{P}}$ as

$$\begin{aligned}
[\mathrm{Pr}_\pi(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_T)]^{16}_{\mathrm{N}} = {} & \mathrm{P_I}(\mathrm{s}_1)\times \\
& \mathrm{P_A}\left(\pi, 1 : \mathrm{a}_1 \mid \mathrm{s}_1\right)\mathrm{P_T}\left(\mathrm{s}_2 \mid \mathrm{a}_1, \mathrm{s}_1\right)\times \\
& \mathrm{P_A}\left(\pi, 2 : \mathrm{a}_2 \mid \mathrm{s}_2\right)\mathrm{P_T}\left(\mathrm{s}_3 \mid \mathrm{a}_2, \mathrm{s}_2\right)\times \\
& \ldots \times \\
& \mathrm{P_A}\left(\pi, T-1 : \mathrm{a}_{T-1} \mid \mathrm{s}_{T-1}\right)\mathrm{P_T}\left(\mathrm{s}_T \mid \mathrm{a}_{T-1}, \mathrm{s}_{T-1}\right),
\end{aligned}$$

or, in more compact [product notation]$^{14}_{\mathrm{N}}$,

$$\mathrm{Pr}_\pi(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_T) = \mathrm{P_I}(\mathrm{s}_1) \times \prod_{t=1}^{T-1} \mathrm{P_A}\left(\pi, t : \mathrm{a}_t \mid \mathrm{s}_t\right)\mathrm{P_T}\left(\mathrm{s}_{t+1} \mid \mathrm{a}_t, \mathrm{s}_t\right).$$

Proposition [3]: /T/rl_theory/P/actcind.

We have that A_t is independent of all previous states and actions given S_t. Notationally, for all $\pi \in \mathbf{\Pi}$, $1 \leq t \leq T-1$, when defined,

$$\Pr_\pi(\mathrm{A}_t \mid \mathrm{S}_t, \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1) = \Pr_\pi(\mathrm{A}_t \mid \mathrm{S}_t).$$

Proof.

Because $\Pr_\pi(\mathrm{A}_t \mid \mathrm{S}_t, \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1)$ is defined, we have that

$$\Pr_\pi(\mathrm{S}_t, \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1) > 0,$$

implying that

$$\Pr_\pi(\mathrm{S}_t) > 0.$$

Therefore, $\Pr_\pi(\mathrm{A}_t \mid \mathrm{S}_t)$ is also defined. Suppose $\mathrm{A}_t = \mathrm{a} \in \mathbf{A}$ and $\mathrm{S}_t = \mathrm{s} \in \mathbf{S}$. [It follows by definition]$^{25}_{\mathrm{D}}$ that

$$\Pr_\pi(\mathrm{A}_t = \mathrm{a} \mid \mathrm{S}_t = \mathrm{s}) = \mathrm{P}_\mathrm{A}(\pi, t : \mathrm{a} \mid \mathrm{s}),$$

so [there are two cases]$^{28}_{\mathrm{D}}$ to consider:

- Case 1: $\Pr_\pi(\mathrm{A}_t = \mathrm{a} \mid \mathrm{S}_t = \mathrm{s}) = 1$.

  [It follows more generally]$^{7}_{\mathrm{P}}$ that, for any random variables

  $$\mathrm{X} \text{ supported on } \mathbf{X}, \mathrm{Y} \text{ supported on } \mathbf{Y}, \mathrm{Z} \text{ supported on } \mathbf{Z},$$

  we have that, when $\Pr(\mathrm{X} \mid \mathrm{Y}, \mathrm{Z})$ is defined,

  $$\Pr(\mathrm{X} \mid \mathrm{Y}) = 1 \implies \Pr(\mathrm{X} \mid \mathrm{Y}, \mathrm{Z}) = 1.$$

- Case 2: $\Pr_\pi(\mathrm{A}_t = \mathrm{a} \mid \mathrm{S}_t = \mathrm{s}) = 0$.

  [It follows more generally]$^{6}_{\mathrm{P}}$ that, for any random variables

  $$\mathrm{X} \text{ supported on } \mathbf{X}, \mathrm{Y} \text{ supported on } \mathbf{Y}, \mathrm{Z} \text{ supported on } \mathbf{Z},$$

  we have that, when $\Pr(\mathrm{X} \mid \mathrm{Y}, \mathrm{Z})$ is defined,

  $$\Pr(\mathrm{X} \mid \mathrm{Y}) = 0 \implies \Pr(\mathrm{X} \mid \mathrm{Y}, \mathrm{Z}) = 0.$$

Proposition [4]: /T/rl_theory/P/npol.

We have that the total number of strategies can be written as

$$|\mathbf{\Pi}| = T \times |\mathbf{S}| \times |\mathbf{A}|.$$

Proof.

This follows from simple combinatorics. Consider an arbitrary strategy

$$\pi = (\pi_1, \ldots, \pi_T).$$

Because each step-strategy $\pi_t : \mathbf{S} \to \mathbf{A}$, there are a total of $|\mathbf{S}| \times |\mathbf{A}|$ distinct π_t possible.

Because each strategy $\pi \in \mathbf{\Pi}$ consists of T such step-strategies π_t, there are a total of

$$|\mathbf{\Pi}| = T \times |\mathbf{S}| \times |\mathbf{A}|$$

distinct π possible.

Proposition [5]: /T/rl_theory/P/pexpr.

We have that for

$$\text{any particular sequence of states } \mathrm{s}_1, \ldots, \mathrm{s}_T \in \mathbf{S} \text{ and actions } \mathrm{a}_1, \ldots, \mathrm{a}_{T-1} \in \mathbf{A}$$

and for

$$\text{any strategy } \pi \in \mathbf{\Pi},$$

we can write

$$\mathrm{Pr}_\pi(\mathrm{S}_1 = \mathrm{s}_1, \mathrm{A}_1 = \mathrm{a}_1, \mathrm{S}_2 = \mathrm{s}_2, \ldots, \mathrm{S}_T = \mathrm{s}_T)$$

as

$$\mathrm{Pr}_\pi(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_T) = \mathrm{P}_\mathrm{I}(\mathrm{s}_1) \times \prod_{t=1}^{T-1} \mathrm{P}_\mathrm{A}\left(\pi, t : \mathrm{a}_t \mid \mathrm{s}_t\right) \mathrm{P}_\mathrm{T}\left(\mathrm{s}_{t+1} \mid \mathrm{a}_t, \mathrm{s}_t\right).$$

Proof.

[It can be shown inductively][11]$_\mathrm{P}$ that for any $1 \leq t \leq T$, for

$$\text{any particular sequence of states } \mathrm{s}_1, \ldots, \mathrm{s}_t \in \mathbf{S} \text{ and actions } \mathrm{a}_1, \ldots, \mathrm{a}_{t-1} \in \mathbf{A}$$

and for

$$\text{any strategy } \pi \in \mathbf{\Pi},$$

we can write

$$\mathrm{Pr}_\pi(\mathrm{S}_1 = \mathrm{s}_1, \mathrm{A}_1 = \mathrm{a}_1, \mathrm{S}_2 = \mathrm{s}_2, \ldots, \mathrm{S}_t = \mathrm{s}_t)$$

as

$$\mathrm{Pr}_\pi(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_t) = \mathrm{P}_\mathrm{I}(\mathrm{s}_1) \times \prod_{t'=1}^{t-1} \mathrm{P}_\mathrm{A}\left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right) \mathrm{P}_\mathrm{T}\left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right),$$

for which the result follows in particular when $t = T$.

Proposition [6]: /T/rl_theory/P/actcind/P/pzerocind.

We have that, for any random variables

$$\mathrm{X} \text{ supported on } \mathbf{X},\ \mathrm{Y} \text{ supported on } \mathbf{Y},\ \mathrm{Z} \text{ supported on } \mathbf{Z},$$

we have that, when $\Pr(\mathrm{X} \mid \mathrm{Y}, \mathrm{Z})$ is defined,

$$\Pr(\mathrm{X} \mid \mathrm{Y}) = 0 \implies \Pr(\mathrm{X} \mid \mathrm{Y}, \mathrm{Z}) = 0.$$

Proof.

Suppose $\mathrm{X} = \mathrm{x} \in \mathbf{X}$, $\mathrm{Y} = \mathrm{y} \in \mathbf{Y}$, and $\mathrm{Z} = \mathrm{z} \in \mathbf{Z}$. Assume that $\Pr(\mathrm{x} \mid \mathrm{y}) = 0$ lo and $\Pr(\mathrm{y}, \mathrm{z}) > 0$ rpg.

We have by definition that

$$\Pr(\mathrm{x} \mid \mathrm{y}) \overset{[\ldots]^{9}_{\mathrm{D}}}{=} \frac{\Pr(\mathrm{x}, \mathrm{y})}{\Pr(\mathrm{y})} = 0,$$

so

$$\Pr(\mathrm{x}, \mathrm{y}) = 0,$$

ljz

from which it follows that

$$\Pr(\mathrm{x}, \mathrm{y}, \mathrm{z}) \overset{\mathtt{ljz},[\ldots]^{8}_{\mathrm{P}}}{=} 0,$$

rjz

so

$$\Pr(\mathrm{x} \mid \mathrm{y}, \mathrm{z}) \overset{[\ldots]^{9}_{\mathrm{D}}}{=} \frac{\Pr(\mathrm{x}, \mathrm{y}, \mathrm{z})}{\Pr(\mathrm{y}, \mathrm{z})} \overset{\mathtt{rjz}}{=} 0.$$

Proposition [7]: /T/rl_theory/P/actcind/P/ponecind.

We have that, for any random variables

$$\mathrm{X} \text{ supported on } \mathbf{X},\ \mathrm{Y} \text{ supported on } \mathbf{Y},\ \mathrm{Z} \text{ supported on } \mathbf{Z},$$

we have that, when $\Pr(\mathrm{X} \mid \mathrm{Y}, \mathrm{Z})$ is defined,

$$\Pr(\mathrm{X} \mid \mathrm{Y}) = 1 \implies \Pr(\mathrm{X} \mid \mathrm{Y}, \mathrm{Z}) = 1.$$

Proof.

Suppose $\mathrm{X} = \mathrm{x} \in \mathbf{X}$, $\mathrm{Y} = \mathrm{y} \in \mathbf{Y}$, and $\mathrm{Z} = \mathrm{z} \in \mathbf{Z}$. Assume that $\Pr(\mathrm{x} \mid \mathrm{y}) = 1$ lo and $\Pr(\mathrm{y}, \mathrm{z}) > 0$ rpg.

By assumption,

$$\Pr(\mathrm{x}, \mathrm{y}) \overset{\texttt{lo},[\ldots]^{9}_{\mathrm{D}}}{=} \Pr(\mathrm{y}) \overset{[\ldots]^{10}_{\mathrm{D}}}{=} \Pr(\mathrm{x}, \mathrm{y}) + \sum_{\mathrm{x}' \in \mathbf{X}, \mathrm{x}' \neq \mathrm{x}} \Pr(\mathrm{x}', \mathrm{y}),$$

so

$$\forall \mathrm{x}' \in \mathbf{X}, \mathrm{x}' \neq \mathrm{x},\ \Pr(\mathrm{x}', \mathrm{y}) = 0,$$

and in particular

$$\forall \mathrm{x}' \in \mathbf{X}, \mathrm{x}' \neq \mathrm{x},\ \Pr(\mathrm{x}', \mathrm{y}, \mathrm{z}) \overset{\texttt{az}}{=} 0.$$

Therefore,

$$\Pr(\mathrm{x}, \mathrm{y}, \mathrm{z}) \underset{\texttt{arm}}{\overset{\texttt{az}}{=}} \Pr(\mathrm{x}, \mathrm{y}, \mathrm{z}) + \sum_{\mathrm{x}' \in \mathbf{X}, \mathrm{x}' \neq \mathrm{x}} \Pr(\mathrm{x}', \mathrm{y}, \mathrm{z}) \underset{\texttt{rm}}{\overset{[\ldots]^{10}_{\mathrm{D}}}{=}} \Pr(\mathrm{y}, \mathrm{z}) \underset{\texttt{gf}}{\overset{\texttt{rpg}}{>}} 0,$$

so

$$\Pr(\mathrm{x} \mid \mathrm{y}, \mathrm{z}) \overset{[\ldots]^{9}_{\mathrm{D}}}{=} \frac{\Pr(\mathrm{x}, \mathrm{y}, \mathrm{z})}{\Pr(\mathrm{y}, \mathrm{z})} \overset{\texttt{arm-rm,gf}}{=} 1.$$

Proposition [8]: /T/rl_theory/P/actcind/P/pzerocind/P/margzimpl.

For random variables

$$X \text{ supported on } \mathbf{X} \text{ and } Y \text{ supported on } \mathbf{Y},$$

we have for all $x \in \mathbf{X}$, $y \in \mathbf{Y}$,

$$\Pr(X = x) = 0 \implies \Pr(X = x, Y = y) = 0.$$

Proof.

Assume that $\Pr(x) = 0$. We have that

lz

$$0 \overset{\text{lz}}{=} \Pr(x) \overset{[\ldots]^{10}_{D}}{=} \sum_{y' \in \mathbf{Y}} \Pr(x, y'),$$

so in particular

$$\Pr(x, y) = 0.$$

Definition [9]: /T/rl_theory/P/actcind/P/ponecind/D/condprob.

Recall from the definition of conditional probability that, for random variables

$$\mathrm{X} \text{ supported on } \mathbf{X} \text{ and } \mathrm{Y} \text{ supported on } \mathbf{Y},$$

we have for all $\mathrm{x} \in \mathbf{X}$, $\mathrm{y} \in \mathbf{Y}$ such that $\Pr(\mathrm{Y} = \mathrm{y}) > 0$:

$$\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) := \frac{\Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y})}{\Pr(\mathrm{Y} = \mathrm{y})},$$

with $\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y})$ undefined if $\Pr(\mathrm{Y} = \mathrm{y}) = 0$.

Definition [10]: /T/rl_theory/P/actcind/P/ponecind/D/margprob.

Recall from the definition of marginal probability that, for random variables

$$\mathrm{X} \text{ supported on } \mathbf{X} \text{ and } \mathrm{Y} \text{ supported on } \mathbf{Y},$$

we have for all $\mathrm{x} \in \mathbf{X}$,

$$\Pr(\mathrm{X} = \mathrm{x}) \coloneqq \sum_{\mathrm{y} \in \mathbf{Y}} \Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}).$$

Proposition [11]: /T/rl_theory/P/pexpr/P/pexprt.

We have that for any $1 \leq t \leq T$, for

$$\text{any particular sequence of states } \mathrm{s}_1, \ldots, \mathrm{s}_t \in \mathbf{S} \text{ and actions } \mathrm{a}_1, \ldots, \mathrm{a}_{t-1} \in \mathbf{A}$$

and for

$$\text{any strategy } \pi \in \mathbf{\Pi},$$

we can write

$$\mathrm{Pr}_\pi(\mathrm{S}_1 = \mathrm{s}_1, \mathrm{A}_1 = \mathrm{a}_1, \mathrm{S}_2 = \mathrm{s}_2, \ldots, \mathrm{S}_t = \mathrm{s}_t)$$

as

$$\mathrm{Pr}_\pi(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_t) = \mathrm{P_I}(\mathrm{s}_1) \times \prod_{t'=1}^{t-1} \mathrm{P_A}\left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right) \mathrm{P_T}\left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right).$$

Proof.

Base case. When $t = 1$, we have [by definition]$^{18}_{\mathrm{D}}$ that $\mathrm{Pr}_\pi(\mathrm{s}_1) = \mathrm{P_I}(\mathrm{s}_1)$.

Inductive hypothesis. Assume that for $t-1$ such that $1 \leq t-1 \leq T-1$, the proposition holds. That is,

$$\mathrm{Pr}_\pi(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}) \underset{\texttt{ih}}{=} \mathrm{P_I}(\mathrm{s}_1) \times \prod_{t'=1}^{t-2} \mathrm{P_A}\left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right) \mathrm{P_T}\left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right).$$

Inductive step. We want to show that the proposition holds for t.

There are two cases to consider:

- Case 1: $\mathrm{Pr}_\pi(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}) \underset{\texttt{co}}{=} 0$. Then,

$$\mathrm{Pr}_\pi(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}) \overset{\texttt{co}}{=} \underset{\texttt{coz}}{0} \overset{[\ldots]^{8}_{\mathrm{P}}}{=} \mathrm{Pr}_\pi(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_t),$$

  and

$$\begin{aligned}
&\mathrm{P_I}(\mathrm{s}_1) \times \prod_{t'=1}^{t-1} \mathrm{P_A}\left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right) \mathrm{P_T}\left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right) \\
&= \mathrm{P_I}(\mathrm{s}_1) \times \prod_{t'=1}^{t-2} \mathrm{P_A}\left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right) \mathrm{P_T}\left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right) \times \\
&\quad \mathrm{P_A}\left(\pi, t-1 : \mathrm{a}_{t-1} \mid \mathrm{s}_{t-1}\right) \times \mathrm{P_T}\left(\mathrm{s}_t \mid \mathrm{a}_{t-1}, \mathrm{s}_{t-1}\right) \\
&\overset{\texttt{ih}}{=} \mathrm{Pr}_\pi(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}) \times \mathrm{P_A}\left(\pi, t-1 : \mathrm{a}_{t-1} \mid \mathrm{s}_{t-1}\right) \times \mathrm{P_T}\left(\mathrm{s}_t \mid \mathrm{a}_{t-1}, \mathrm{s}_{t-1}\right) \\
&\overset{\texttt{co}\ \texttt{coz}}{=} 0 = \mathrm{Pr}_\pi(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_t).
\end{aligned}$$

- Case 2: $\mathrm{Pr}_\pi(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}) \underset{\texttt{ct}}{>} 0$. Then,

$$\underset{\texttt{daf}}{\mathrm{Pr}_\pi(\mathrm{a}_{t-1} \mid \mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{t-1}) \text{ is defined}}^{\texttt{ct},[\ldots]^{9}_{\mathrm{D}}}, \underset{\texttt{da}}{\mathrm{Pr}_\pi(\mathrm{a}_{t-1} \mid \mathrm{s}_{t-1}) \text{ is defined}}^{\texttt{ct},[\ldots]^{12}_{\mathrm{P}},[\ldots]^{9}_{\mathrm{D}}},$$

and we have

$$\Pr_\pi(a_{t-1} \mid s_1, a_1, s_2, \ldots, s_{t-1}) \overset{\texttt{daf},[\ldots]^{3}_{\mathrm{P}}}{\underset{\texttt{cia}}{=}} \Pr_\pi(a_{t-1} \mid s_{t-1}) \overset{\texttt{da},[\ldots]^{25}_{\mathrm{D}}}{\underset{\texttt{ad}}{=}} \mathrm{P_A}\left(\pi, t-1 : a_{t-1} \mid s_{t-1}\right).$$

[There are two cases]$^{28}_{\mathrm{D}}$ to consider:

– Case 2.1: $\mathrm{P_A}\left(\pi, t-1 : a_{t-1} \mid s_{t-1}\right) \underset{\texttt{cto}}{=} 1$. Then,

$$\begin{aligned}
\Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}, a_{t-1}) &\overset{\texttt{daf}}{\underset{\texttt{md}}{=}} \Pr_\pi(a_{t-1} \mid s_1, a_1, s_2, \ldots, s_{t-1}) \times \Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}) \\
&\overset{\texttt{cia-ad}}{=} \mathrm{P_A}\left(\pi, t-1 : a_{t-1} \mid s_{t-1}\right) \times \Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}) \\
&\overset{\texttt{cto}}{=} \Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}) \\
&\overset{\texttt{ct}}{>} 0.
\end{aligned}$$

Therefore,

$$\Pr_\pi(s_t \mid a_{t-1}, s_1, a_1, s_2, \ldots, s_{t-1}) \text{ is defined}, \Pr_\pi(s_t \mid a_{t-1}, s_{t-1}) \text{ is defined}$$

and we have

$$\Pr_\pi(s_t \mid a_{t-1}, s_1, a_1, s_2, \ldots, s_{t-1}) \overset{\texttt{dsf},[\ldots]^{24}_{\mathrm{D}}}{\underset{\texttt{cis}}{=}} \Pr_\pi(s_t \mid a_{t-1}, s_{t-1}) \overset{\texttt{ds},[\ldots]^{33}_{\mathrm{D}}}{\underset{\texttt{sd}}{=}} \mathrm{P_T}\left(s_t \mid a_{t-1}, s_{t-1}\right).$$

Therefore,

$$\begin{aligned}
\Pr_\pi(s_1, a_1, s_2, \ldots, s_t) &\overset{\texttt{daf,dsf}}{=} \Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}) \times \Pr_\pi(a_{t-1} \mid s_1, a_1, s_2, \ldots, s_{t-1}) \times \\
&\qquad \Pr_\pi(s_t \mid a_{t-1}, s_1, a_1, s_2, \ldots, s_{t-1}) \\
&\overset{\texttt{cia,cis}}{=} \Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}) \times \Pr_\pi(a_{t-1} \mid s_{t-1}) \times \Pr_\pi(s_t \mid a_{t-1}, s_{t-1}) \\
&\overset{\texttt{ih,ad,sd}}{=} \mathrm{P_I}(s_1) \times \prod_{t'=1}^{t-1} \mathrm{P_A}\left(\pi, t' : a_{t'} \mid s_{t'}\right) \mathrm{P_T}\left(s_{t'+1} \mid a_{t'}, s_{t'}\right).
\end{aligned}$$

– Case 2.2: $\mathrm{P_A}\left(\pi, t-1 : a_{t-1} \mid s_{t-1}\right) \underset{\texttt{ctt}}{=} 0$. Then,

$$\begin{aligned}
\Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}, a_{t-1}) &\overset{\texttt{daf}}{=} \Pr_\pi(a_{t-1} \mid s_1, a_1, s_2, \ldots, s_{t-1}) \times \Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}) \\
&\overset{\texttt{cia-ad}}{=} \mathrm{P_A}\left(\pi, t-1 : a_{t-1} \mid s_{t-1}\right) \times \Pr_\pi(s_1, a_1, s_2, \ldots, s_{t-1}) \\
&\overset{\texttt{ctt}}{=} 0.
\end{aligned}$$

Therefore,

$$\mathrm{Pr}_\pi(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_t) = 0,$$

cttz

so

$$\mathrm{Pr}_\pi(\mathrm{s}_1, \mathrm{a}_1, \mathrm{s}_2, \ldots, \mathrm{s}_t) \overset{\texttt{cttz}}{=} 0 \overset{\texttt{ctt}}{=} \mathrm{P_I}(\mathrm{s}_1) \times \prod_{t'=1}^{t-1} \mathrm{P_A}\left(\pi, t' : \mathrm{a}_{t'} \mid \mathrm{s}_{t'}\right) \mathrm{P_T}\left(\mathrm{s}_{t'+1} \mid \mathrm{a}_{t'}, \mathrm{s}_{t'}\right).$$

Proposition [12]: /T/rl_theory/P/pexpr/P/pexprt/P/marggimpl.

For random variables

$$\mathrm{X} \text{ supported on } \mathbf{X} \text{ and } \mathrm{Y} \text{ supported on } \mathbf{Y},$$

we have for all $\mathrm{x} \in \mathbf{X}$, $\mathrm{y} \in \mathbf{Y}$,

$$\Pr(\mathrm{X} = \mathrm{x}, \mathrm{Y} = \mathrm{y}) > 0 \implies \Pr(\mathrm{X} = \mathrm{x}) > 0.$$

Proof.

Assume that $\Pr(\mathrm{x}, \mathrm{y}) > 0$. We have that

lg

$$\Pr(\mathrm{x}) \overset{[\cdots]^{10}_{\mathrm{D}}}{=} \sum_{\mathrm{y}' \in \mathbf{Y}} \Pr(\mathrm{x}, \mathrm{y}') \geq \Pr(\mathrm{x}, \mathrm{y}) \overset{\text{lg}}{>} 0.$$

Note [13]: /T/rl_theory/N/paramprob.

We use the notation $\Pr(\cdot)$ and $\mathrm{P}(\cdot)$ to distinguish between a probability and a "probability parameterization", respectively.

A probability parameterization is simply a function mapping to probability values, and is defined for every possible input. The form of the inputs are implicit in its definition.

A probability, however, may or may not be defined, specifically when it is a conditional probability.

When we correspond a probability with a probability parameterization, we almost always mean to say that, for a particular input, they are equal when the probability is defined.

For example, consider the random variables X, Y, supported on $\mathbf{X}$ and $\mathbf{Y}$ respectively. For $\mathrm{x} \in \mathbf{X}$, $\mathrm{y} \in \mathbf{Y}$, if we correspond $\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y})$ with $\mathrm{P}(\mathrm{x} \mid \mathrm{y})$, we mean to say that when $\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y})$ is defined (that is, when $\Pr(\mathrm{Y} = \mathrm{y}) > 0$),

$$\Pr(\mathrm{X} = \mathrm{x} \mid \mathrm{Y} = \mathrm{y}) = \mathrm{P}(\mathrm{x} \mid \mathrm{y}).$$

Note [14]: /T/rl_theory/N/prodnot .

Consider an arbitrary function $f(\cdot)$. The notation

$$\prod_{x=a}^{b} f(x),$$

where a, b are integers, denotes the product $f(a) \times f(a+1) \times \cdots \times f(b-1) \times f(b)$ when $a \leq b$, and 1 otherwise. That is,

$$\prod_{x=a}^{b} f(x) = \begin{cases} f(a) \times f(a+1) \times \cdots \times f(b-1) \times f(b) & a \leq b \\ 1 & a > b. \end{cases}$$

Note [15]: /T/rl_theory/N/implgen.

Consider the random variable X supported on $\mathbf{X}$. When we write X by itself in a probability equation, we mean to say that

the equation applies when X is replaced with $X = x$ for all $x \in \mathbf{X}$.

This applies more generally to multiple random variables, for every combination of assignments.

Note [16]: /T/rl_theory/N/implasn.

Consider the random variable X supported on $\mathbf{X}$. When we write

$$\Pr(X = x)$$

for some $x \in \mathbf{X}$, we associate the random variable X with the assignment x.

We may later, for brevity, write this same quantity as

$$\Pr(x).$$

It should always be clear from the context what random variable is being assigned. This is referred to as "implied assignment."

This applies more generally to multiple random variables, for every combination of assignments.

Note [17]: /T/rl_theory/N/fnnot .

We use the notation

$$f : X \to Y$$

to denote a function with domain X and range Y; that is,

$$\text{for all } x \in X, \text{ there is some } y \in Y \text{ such that } f(x) = y.$$

Definition [18]: /T/rl_theory/D/ind.

We have by definition that for all $\pi \in \mathbf{\Pi}$, $\mathrm{s} \in \mathbf{S}$,

$$\mathrm{Pr}_{\pi}(\mathrm{S}_1 = \mathrm{s}) = \mathrm{P}_{\mathrm{I}}(\mathrm{s}).$$

Definition [19]: /T/rl_theory/D/rewards.

We have by definition that

$$R : \mathbf{S} \to \mathbb{R} := \text{mapping from states to returns.}$$

Definition [20]: /T/rl_theory/D/pstr.

We have by definition that

$$\Pr_\pi(\cdot) \coloneqq \text{the probability of an event when the agent follows a strategy } \pi \in \mathbf{\Pi}.$$

Definition [21]: /T/rl_theory/D/steplim.

We have by definition that

$$T \coloneqq \text{the step limit, where } T \geq 1.$$

Definition [22]: /T/rl_theory/D/str.

We have by definition that

$$\pi = (\pi_1, \ldots, \pi_{T-1}) \coloneqq \text{a strategy, where each “step-strategy” } \pi_t : \mathbf{S} \to \mathbf{A}.$$

Definition [23]: /T/rl_theory/D/inddef.

We have by definition that for any $s \in \mathbf{S}$,

$$P_I(s) := \text{the initial distribution probability of starting in state } s.$$

Definition [24]: /T/rl_theory/D/trdcind.

We have by definition that S_t is independent of all previous states and actions given A_{t-1} and S_{t-1}. Notationally, for all $\pi \in \mathbf{\Pi}$, $2 \leq t \leq T$, when defined,

$$\Pr_\pi(\mathrm{S}_t \mid \mathrm{A}_{t-1}, \mathrm{S}_{t-1}, \ldots, \mathrm{A}_1, \mathrm{S}_1) = \Pr_\pi(\mathrm{S}_t \mid \mathrm{A}_{t-1}, \mathrm{S}_{t-1}).$$

Definition [25]: /T/rl_theory/D/pact.

We have by definition that, for all $\pi \in \mathbf{\Pi}$, $\mathrm{s} \in \mathbf{S}$, $\mathrm{a} \in \mathbf{A}$, $1 \leq t \leq T - 1$, when defined,

$$\mathrm{Pr}_{\pi}(\mathrm{A}_t = \mathrm{a} \mid \mathrm{S}_t = \mathrm{s}) = \mathrm{P}_{\mathrm{A}}(\pi, t : \mathrm{a} \mid \mathrm{s}).$$

Definition [26]: /T/rl_theory/D/states.

We have by definition that

$$\mathbf{S} := \text{the set of states},$$

where $\mathbf{S}$ is finite.

Definition [27]: /T/rl_theory/D/rvat.

We have by definition that for all $1 \leq t \leq T - 1$,

$$\mathrm{A}_t \coloneqq \text{random variable for the action at step } t.$$

Definition [28]: /T/rl_theory/D/pactdef.

We have by definition that for any $\mathrm{s} \in \mathbf{S}$, $\mathrm{a} \in \mathbf{A}$, $\pi \in \mathbf{\Pi}$, $1 \leq t \leq T - 1$,

$$\mathrm{P}_{\mathrm{A}}\left(\pi, t : \mathrm{a} \mid \mathrm{s}\right) \coloneqq \begin{cases} 1 & \pi_t(\mathrm{s}) = \mathrm{a} \\ 0 & \pi_t(\mathrm{s}) \neq \mathrm{a}. \end{cases}$$

Definition [29]: /T/rl_theory/D/rvst.

We have by definition that for all $1 \leq t \leq T$,

$$\mathrm{S}_t \coloneqq \text{random variable for the state at step } t.$$

Definition [30]: /T/rl_theory/D/actions.

We have by definition that

$$\mathbf{A} \coloneqq \text{the set of actions},$$

where $\mathbf{A}$ is finite.

Definition [31]: /T/rl_theory/D/trddef.

We have by definition that for any $\mathrm{s}, \mathrm{s}' \in \mathbf{S}$, $\mathrm{a} \in \mathbf{A}$,

$\mathrm{P}_{\mathrm{T}}(\mathrm{s} \mid \mathrm{a}, \mathrm{s}') :=$ the transition distribution probability of going to state s after action a in state s'.

Definition [32]: /T/rl_theory/D/estr.

We have by definition that

$$\mathbb{E}_\pi[\cdot] \coloneqq \text{the expectation of a quantity when the agent follows a strategy } \pi \in \mathbf{\Pi}.$$

Definition [33]: /T/rl_theory/D/trd.

We have by definition that for all $\pi \in \mathbf{\Pi}$, $\mathrm{s}, \mathrm{s}' \in \mathbf{S}$, $\mathrm{a} \in \mathbf{A}$, $2 \leq t \leq T$, when defined,

$$\mathrm{Pr}_{\pi}(\mathrm{S}_t = \mathrm{s} \mid \mathrm{A}_{t-1} = \mathrm{a}, \mathrm{S}_{t-1} = \mathrm{s}') = \mathrm{P}_{\mathrm{T}}\left(\mathrm{s} \mid \mathrm{a}, \mathrm{s}'\right).$$

Definition [34]: /T/rl_theory/D/strs.

We have by definition that

$$\mathbf{\Pi} \coloneqq \text{the set of all strategies } \pi.$$